

L6: Rich-Club Set of Nodes

A common approach for the detection of core-periphery structure is referred to as the **rich-club** of a network.

The metaphor behind this concept comes from social networks: the very rich people are few and they have a large number of acquaintances, which include the rest of those few very rich people. So those few very rich people form a highly clustered small group of nodes (*the “rich-club”*) that is also very well-connected with the rest of the network.

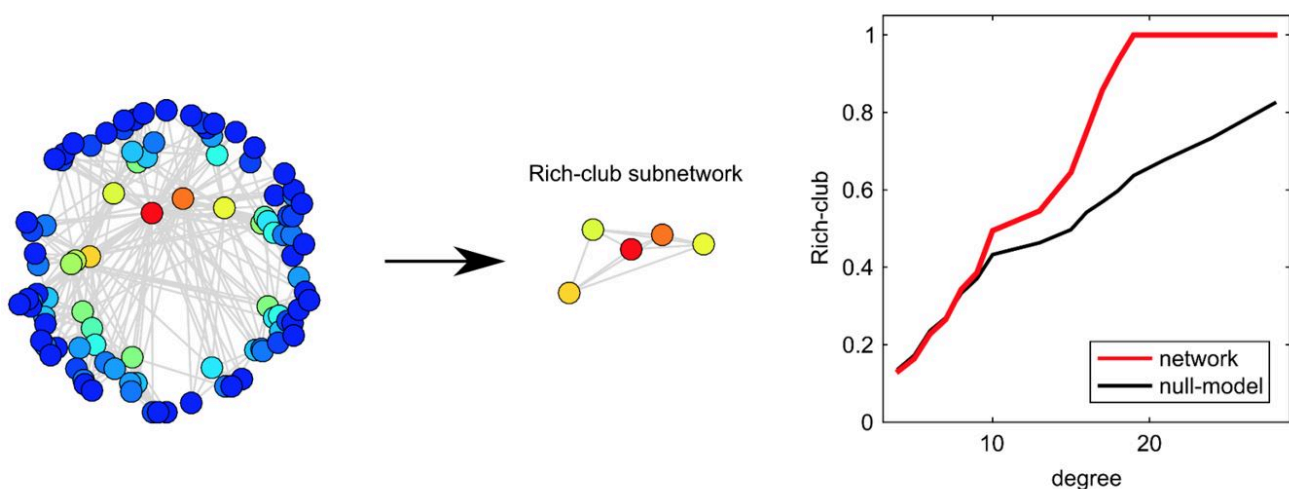
To define this notion for general (*but undirected*) networks, suppose that *the number of nodes with degree greater than k is n_k* . Let e_k be the number of edges between those nodes. The maximum number of possible edges between those nodes is $\frac{n_k(n_k-1)}{2}$. The “rich-club coefficient” for degree k is defined as:

$$\phi(k) = \frac{2 e_k}{n_k (n_k - 1)}$$

and it quantifies the density of the connections between nodes of degree greater than k .

How can we tell however whether the value of $\phi(k)$ is statistically significant for a given k ? It could be that even randomly wired networks have about the same value of $\phi(k)$, at least for some degree values. To do so, we also generate an ensemble of random networks with the same number of nodes and degree distribution (*as we did in Lesson-4 for the detection of network motifs*). These random networks represent our **null model**. We can then compute the average value of $\phi(k)$ for each k , averaging across all random networks.

If the rich-club coefficient for degree k in the given network is much larger than the corresponding coefficient in the null model, we can mark that value of k as statistically significant for the existence of a rich-club. If there is at least one such statistically significant value of k , we conclude that the network includes a rich-club -- the set of nodes with degree greater than k . If there are multiple such values of k , the rich-club nodes can be identified based on the value of k for which we have the largest difference between the rich-club coefficient in the real network versus the null model (even though there is some variation about this in the literature).



The visualization shows a synthetic network (left). The plot at the right shows the rich-club coefficient as a function of the degree k for the given network (red) as well as for the null model. The value of k for which we are most confident that a rich-club exists is $k=19$. There are five nodes with degree greater than 19 –they are shown (*separately than the rest of the network*) at the center of the visualization. Note that the corresponding rich-club coefficient is 1, which means that these five nodes form a clique. If we had chosen a higher value of k , the rich-club would only consist of a subset of these five nodes.

Food For Thought

Think about the similarities and differences between the following notions:

a) the rich-club set of nodes (*if it exists*)

b) the maximum order k-core group (*this set of nodes always exists*)

c) a group of nodes that can be described as **hubs** because *their degree is higher than the average degree plus 3 standard deviations (they may or may not exist)*.